

Deep Xi as a Front-End for Robust Automatic Speech Recognition

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Abstract—Currently, deep learning approaches to speech enhancement are most commonly used as front-ends for robust automatic speech recognition (ASR). A recently proposed deep learning approach to *a priori* SNR estimation, called Deep Xi, was able to produce enhanced speech at a higher quality and intelligibility than recent deep learning approaches to speech enhancement. Motivated by this, we investigate Deep Xi as a front-end for robust ASR. Deep Xi is evaluated using real-world non-stationary and coloured noise sources at multiple SNR levels. Our experimental investigation shows that Deep Xi as a front-end is able to produce a lower word error rate than recent deep learning approaches to speech enhancement. The results presented in this work show that Deep Xi is a viable front-end, and is able to significantly increase the robustness of an ASR system.

Availability: Deep Xi is available at <https://github.com/anicolson/DeepXi>.

Index Terms: Robust speech recognition, front-end, speech enhancement, pre-processing, Deep Xi

I. INTRODUCTION

Recently, an automatic speech recognition (ASR) system was able to achieve human parity on the Switchboard speech recognition task [1]. This milestone demonstrates how far ASR research has come in 68 years [2]. An example of a modern ASR system is Deep Speech [3] — an end-to-end ASR system that uses a bidirectional recurrent neural network (BRNN) as its acoustic model [4]. Despite their accuracy on ideal conditions, modern ASR systems are still susceptible to performance degradation when environmental noise is present. A popular method to increase the robustness of an ASR system is to use a front-end to pre-process noise corrupted speech. As the goal of a front-end is to perform noise suppression, a speech enhancement algorithm is typically employed.

Deep learning approaches to speech enhancement are currently the leading front-ends in the literature [5]. They have the ability to produce enhanced speech that is highly intelligible — an important attribute for ASR [6]. One prominent example is the long short-term memory network ideal ratio mask (LSTM-IRM) estimator [7], which can produce intelligible enhanced speech independent of the speaker. Another prominent example is the fully-connected neural network that employs multi-objective learning and binary mask post-processing to estimate the clean speech log-power spectra (LPS) (Xu2017) [8]. Generative adversarial networks (GAN) are also commonly used for speech enhancement. The most prominent example utilises an encoder-decoder convolutional neural network (CNN) as the generator and the discriminator, called SEGAN [9].

Previous generation front-ends include minimum mean-square error (MMSE) approaches to speech enhancement and missing data approaches. MMSE approaches, such as the MMSE short-time spectral amplitude (MMSE-STSA) estimator, rely on the accurate estimation of the *a priori* SNR [10]. However, previous *a priori* SNR estimators, like the decision-directed (DD) approach, introduce a tracking delay and a large amount of bias [10]. Cluster-based reconstruction is a prominent missing data approach, which reconstructs the unreliable spectral components (components with an *a priori* SNR of 0 dB or less [11]) based on their statistical relationship to the reliable components [12].

A deep learning approach to *a priori* SNR estimation was recently proposed, called Deep Xi [13]. It is able to produce an *a priori* SNR estimate with negligible bias, and exhibits no tracking delay. Deep Xi enabled MMSE approaches to achieve higher quality and intelligibility scores than recent deep learning approaches to speech enhancement. As Deep Xi is able to produce more intelligible enhanced speech than deep learning approaches to speech enhancement, we propose that Deep Xi can further outperform them when used as a front-end for robust ASR.

Here, Deep Xi as a front-end is evaluated using real-world non-stationary and coloured noise sources at multiple SNR levels. Deep Speech is used to evaluate each of the front-ends. Deep Speech is suitable for front-end evaluation as it is trained on multiple clean speech corpora and has no implemented front- or back-end techniques. The paper is organised as follows: an overview of Deep Xi is given in Section II; the proposed front-end is presented in Section III; the experiment setup is described in Section IV, including a description of each front-end; the results and discussion are presented in Section V; conclusions are drawn in Section VI.

II. OVERVIEW OF DEEP XI

In [13], the Deep Xi framework for estimating the *a priori* SNR was proposed. The *a priori* SNR is given by:

$$\xi[l, k] = \frac{\lambda_s[l, k]}{\lambda_d[l, k]}, \quad (1)$$

where l is the time-frame sample, k is the discrete-frequency bin, $\lambda_s[l, k] = E\{|S[l, k]|^2\}$ is the variance of the clean speech spectral component, and $\lambda_d[l, k] = E\{|D[l, k]|^2\}$ is the variance of the noise spectral component. During training, the clean speech and noise of the noisy speech are available. This

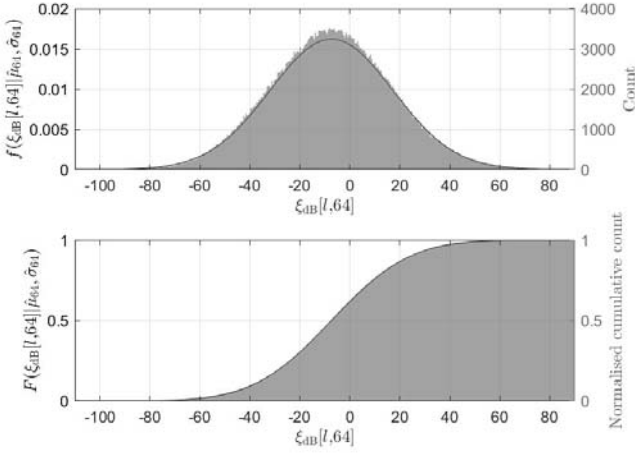


Fig. 1. Histogram of $\xi_{dB}[l, 64]$ over a sample of the training set and the probability density function (PDF) of $\xi_{dB}[l, 64]$ ($f(\xi_{dB}[l, 64]|\mu_{64}, \sigma_{64})$) (top). Cumulative histogram of $\xi_{dB}[l, 64]$ over a sample of the training set and the CDF of $\xi_{dB}[l, 64]$ ($F(\xi_{dB}[l, 64]|\mu_{64}, \sigma_{64})$) (bottom). Both μ_{64} and σ_{64} are estimated from the sample of the training set, as described in Subsection IV-B.

allows the instantaneous case of the *a priori* SNR to be used as the training target. To compute the instantaneous *a priori* SNR, $\lambda_s[l, k]$ and $\lambda_d[l, k]$ in Equation (1) are replaced with the squared magnitude of the clean speech and noise spectral components, respectively.

The observation and target of a training example for a deep neural network (DNN) in the Deep Xi framework is as follows:¹

- **Observation:** noisy speech short-time magnitude spectrum (STMS) of the l^{th} time-frame, $|\mathbf{X}_l|$.
- **Target:** mapped *a priori* SNR for the l^{th} time-frame, $\bar{\xi}_l$.

The mapped *a priori* SNR is a mapped version of the instantaneous *a priori* SNR. The instantaneous *a priori* SNR is mapped to the interval $[0, 1]$ in order to improve the rate of convergence of the used stochastic gradient descent algorithm [13]. The cumulative distribution function (CDF) of $\xi_{dB}[l, k] = 10 \log_{10}(\xi[l, k])$ is used as the map. As shown in Figure 1 (top) and 2, the distribution of $\xi_{dB}[l, k]$ for each discrete-frequency bin follows a normal distribution. It is thus assumed that $\xi_{dB}[l, k]$ is distributed normally with mean μ_k and variance σ_k^2 : $\xi_{dB}[l, k] \sim \mathcal{N}(\mu_k, \sigma_k^2)$. Thus, the mapped *a priori* SNR is found by applying the normal CDF to $\xi_{dB}[l, k]$:

$$\bar{\xi}[l, k] = F(\xi_{dB}[l, k]|\mu_k, \sigma_k) = \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{\xi_{dB}[l, k] - \mu_k}{\sigma_k \sqrt{2}} \right) \right], \quad (2)$$

where μ_k and σ_k^2 are found over a sample of the training set, as described in Subsection IV-B. The map for $k = 64$ is depicted in Figure 1 (bottom).

¹In this work, $|\mathbf{X}[l, k]|$ specifies the k^{th} component of the l^{th} time-frame, $|\mathbf{X}_l|$ specifies $k = \{0, \dots, K-1\}$ of the l^{th} time-frame (where K is the number of discrete-frequency bins), and $|\mathbf{X}|$ specifies $k = \{0, \dots, K-1\}$ and $L = \{1, \dots, L\}$ (where L is the number of time-frames).

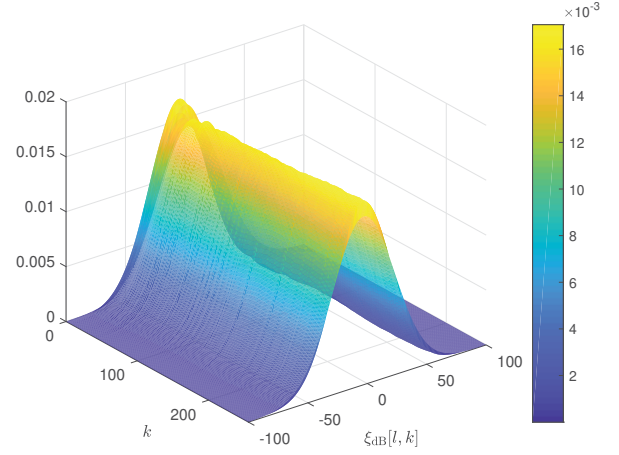


Fig. 2. The distribution of $\xi_{dB}[l, k]$, for each discrete-frequency bin, k . For each k , μ_k and σ_k are estimated from the sample of the training set, as described in Subsection IV-B.

During inference, the *a priori* SNR estimate, $\hat{\xi}[l, k]$ is found from $\hat{\xi}[l, k]$ as

$$\hat{\xi}[l, k] = 10 \left((\sigma_k \sqrt{2} \operatorname{erf}^{-1}(2\hat{\xi}[l, k] - 1) + \mu_k) / 10 \right). \quad (3)$$

The *a priori* SNR estimate can then be used by an MMSE approach to estimate the clean speech STMS:

$$|\hat{S}[l, k]| = G[l, k] |X[l, k]|, \quad (4)$$

where $G(\cdot)$ is an MMSE approach gain function and $\hat{\gamma}[l, k]$ is the *a posteriori* SNR. As in [13], the maximum-likelihood (ML) estimate of the *a posteriori* SNR is used [14]:

$$\hat{\gamma}[l, k] = \xi[l, k] + 1. \quad (5)$$

One example of an MMSE approach gain function is the MMSE log-spectral amplitude (MMSE-LSA) estimator [15]:

$$G_{\text{MMSE-LSA}}[l, k] = \frac{\xi[l, k]}{\xi[l, k] + 1} \exp \left\{ \frac{1}{2} \int_{v[l, k]}^{\infty} \frac{e^{-t}}{t} dt \right\}, \quad (6)$$

where $v[l, k] = \gamma[l, k] \xi[l, k] / (\xi[l, k] + 1)$. Other MMSE approaches include the MMSE-STSA estimator [10], Wiener filter (WF) [16], constrained WF (cWF) [16], and square-root WF (SRWF) [17].

III. DEEP XI AS A FRONT-END

In this section, Deep Xi as a front-end for robust ASR is described. Project DeepSpeech² is used as the back-end of the ASR system in this work. It is an open source implementation of the Deep Speech ASR system [3]. It uses 26 mel-frequency cepstral coefficients (MFCCs) as its input. The process of finding the hypothesis transcription, H , is illustrated in Figure 3. The process includes the following four steps:

²Project DeepSpeech is available at: <https://github.com/mozilla/DeepSpeech> (model 0.1.1 is used).

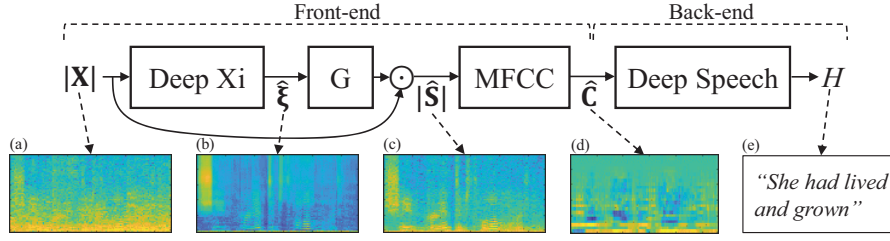


Fig. 3. Deep Xi as a front-end for robust ASR. The noisy speech magnitude spectra, $|X|$, is shown in (a). Deep Xi estimates the *a priori* SNR, $\hat{\xi}$, as shown in (b). An MMSE approach is next used to compute a gain function, $G(\hat{\xi}, \hat{\gamma})$, which is then multiplied element-wise with $|X|$ to produce the estimated clean speech magnitude spectra, $|\hat{S}|$, as shown in (c). MFCCs are then computed, producing the estimated clean speech cepstra, $\hat{C}$, as shown in (d). Finally, the hypothesis transcript, H , is found using Deep Speech, as shown in (e).

- 1) The *a priori* SNR estimate, $\hat{\xi}$, of the noisy speech magnitude spectra, $|X|$, is found using $\hat{\xi} = \text{Deep Xi}(|X|)$.
- 2) $\hat{\xi}$ is then used to estimate clean speech magnitude spectra, $|\hat{S}|$, using Equations (4) and (5).
- 3) The input to Deep Speech is the estimated clean speech cepstra, $\hat{C}$, where $\hat{C}$ is computed from $|\hat{S}|$: $\hat{C} = \text{MFCC}(|\hat{S}|)$, as in [18].
- 4) Deep Speech computes the hypothesised transcript, H , from $\hat{C}$: $H = \text{Deep Speech}(\hat{C})$.

Deep Xi attempts to match the observed noisy speech to the conditions experienced by Deep Speech during training, i.e. the unobserved clean speech is estimated from the noisy speech. Steps one, two, and three form the front-end of the robust ASR system.

IV. EXPERIMENT SETUP

A. Training set

The *train-clean-100* set from the Librispeech corpus [19], the CSTR VCTK corpus [20], and the *si** and *sx** training sets from the TIMIT corpus [21] are used to form the training and validation set (74 250 clean speech recordings). 5% of the clean speech recordings (3 713) are randomly selected and used as the validation set, leaving 70 537 for the training set. The training set includes 138.4 hours of clean speech, while the validation set includes 7.4 hours of clean speech. The 2 382 noise recordings adopted in [13], along with the clean speech recordings are used to create the noisy speech for the training and validation set. All of the clean speech and noise recordings are single-channel, with a sampling frequency of 16 kHz (those with a higher sampling frequency are downsampled). The noisy speech for the validation set is created by corrupting each clean speech recording of the validation set with a random section of a randomly selected noise recording at a randomly selected SNR level from the set $\{-10, -5, 0, 5, 10, 15, 20\}$ (dB). The noisy speech for each training mini-batch is formed by corrupting each clean speech recording selected for the mini-batch with a random section of a randomly selected noise recording at a randomly selected SNR level from the set $\{-10, -5, 0, 5, 10, 15, 20\}$ (dB).

B. Training set sample

The sample mean and standard deviation of $\xi_{\text{dB}}[l, k]$ for $k = \{0, \dots, K-1\}$ are found over 1 250 noisy speech signals created from the clean speech and noise recordings from Subsection IV-A. The 1 250 noisy speech signals are created from 250 randomly selected (without replacement) clean speech recordings that are mixed with random sections of randomly selected (without replacement) noise recordings at five different SNR levels: $\{-5, 0, 5, 10, 15\}$ dB.

C. Test set

Four noise sources are included in the test set: *voice babble*, *F16*, and *factory* from the RSG-10 noise dataset [22] and *street music* (recording no. 26 270) from the Urban Sound dataset [23]. 25 clean speech recordings are randomly selected (without replacement) from the *test-clean* set of the Librispeech corpus [19] for each of the four noise recordings. To generate the noisy speech, a random section of the noise recording is mixed with the clean speech at the following SNR levels: $\{-5, 0, 5, 10, 15\}$ (dB). This creates a test set of 500 noisy speech signals. The noisy speech is single-channel, with a sampling frequency of 16 kHz.

D. Front-ends

The configuration of each front-end is described here:

Deep Xi-ResLSTM & Deep Xi-ResBiLSTM: A residual LSTM (ResLSTM) network and a residual bidirectional LSTM (ResBiLSTM) network were used within the Deep Xi framework, as shown in Figure 4. The hyperparameters for the ResLSTM and ResBiLSTM networks in [13] are employed. The ResLSTM network, as shown in Figure 4 (top), consists of five residual blocks, with each block containing an LSTM cell [24], $\mathbf{F}$, with a cell size of 512. The residual connection is from the input of the residual block to after the LSTM cell activation. $\mathbf{FC}$ is a fully-connected layer with 512 nodes that employs layer normalisation followed by ReLU activation [25]. The output layer, $\mathbf{O}$, is a fully-connected layer with sigmoidal units. The ResBiLSTM network, as shown in Figure 4 (bottom), is identical to the ResLSTM network, except that the residual blocks include both a forward and backward LSTM cell ($\mathbf{F}$ and $\mathbf{B}$, respectively) [4], each with a cell size of 512. The residual connection is applied from the input of

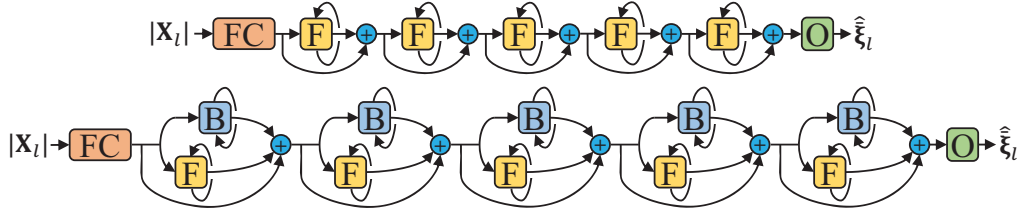


Fig. 4. Deep Xi-ResLSTM (top) and Deep Xi-ResBiLSTM (bottom).

the residual block to after the summation of the forward and backward cell activations. The ResLSTM and ResBiLSTM networks are trained on the training set described in Subsection IV-A for 10 epochs. Cross-entropy is used as the loss function. The *Adam* algorithm [26] with default hyperparameters is used for stochastic gradient descent optimisation. A mini-batch size of 10 noisy speech signals is used.

SEGAN: The trained model available at <https://github.com/santi-pdp/segan> is retrained using the training set described in Subsection IV-A for 50 epochs using the available implementation.

LSTM-IRM: The LSTM-IRM estimator from [7] is trained on the training set described in Subsection IV-A for 10 epochs. The noisy speech STMS is used as the input. The *Adam* algorithm [26] with default hyperparameters is used for stochastic gradient descent optimisation. A mini-batch size of 10 noisy speech signals is used.

Xu2017: The trained model available at <https://github.com/yongxuUSTC/DNN-for-speech-enhancement> is used.

Cluster-based reconstruction: A diagonal covariance Gaussian mixture model (GMM) consisting of 128 clusters is trained using the k-means++ algorithm [27], and the expectation-maximisation algorithm [28]. 7500 randomly selected clean speech recordings from the training set are used for training. A BRNN ideal binary mask (IBM) estimator [29], which is trained for 10 epochs using the training set, is used to identify the unreliable spectral components [12].

DD: The MMSE-based noise power spectral density (PSD) estimator from [30] is used by the DD approach *a priori* SNR estimator [10]. The DD approach estimates the *a priori* SNR for the MMSE-STSA estimator [10].

E. Signal processing

The following hyperparameters were used to compute the inputs used by the DD approach, cluster-based reconstruction, the LSTM-IRM estimator, Deep Xi-ResLSTM, and Deep Xi-ResBiLSTM. A sampling frequency of 16 kHz is used. The Hamming window function is used for analysis, with a frame length of 32 ms (512 discrete-time samples) and a frame shift of 16 ms (256 discrete-time samples). For each frame of noisy speech, the 257-point single-sided magnitude spectrum is computed, which includes both the DC frequency component and the Nyquist frequency component.

F. Word Error Rate Percentage

The word error rate percentage, WER%, was used to evaluate the performance of the front-end techniques. The WER% is calculated by

$$\text{WER}\% = 100 \times \frac{\mathcal{D}(H, R)}{N}, \quad (7)$$

where H is the hypothesis transcript, R is the reference transcript, and N number of words in R . $\mathcal{D}(H, R)$ is the Levenshtein distance between H , and R .

V. RESULTS AND DISCUSSION

The performance of Deep Xi utilising different MMSE approaches is shown in Table I. The SRWF attained the lowest average word error rate percentage (WER%) over all conditions at 37.07%, marginally outperforming the MMSE-STSA estimator (37.09%), and the cWF (37.27%). Although the cWF was not able to achieve the lowest average WER%, it was able to attain the lowest WER% for most individual conditions, especially for *street music*. The performance of the WF was the worst amongst the MMSE approaches, with a WER% of 40.03% over all conditions. It did however perform best for two conditions, *voice babble* at -5 dB, and *factory* at 15 dB.

In Table II, Deep Xi-ResLSTM and Deep Xi-ResBiLSTM are compared to both current and previous generation front-ends using Deep Speech. The current front-ends include Xu2017 [8], an LSTM-IRM estimator [7], and SEGAN [9]. The previous generation front-ends include the DD approach, and cluster-based reconstruction. As it attained the lowest average WER% in Table I, the SRWF is used by Deep Xi-ResLSTM and Deep Xi-ResBiLSTM.

Deep Xi-ResBiLSTM demonstrates a significant performance improvement, with an average WER% reduction of 9.3% over all conditions when compared to SEGAN. Deep Xi-ResLSTM also demonstrated a high performance, with an average WER% reduction of 3.4% over all conditions when compared to SEGAN. As shown in [6], more intelligible speech improves speech recognition performance. In [13], Deep Xi-ResLSTM and Deep Xi-ResBiLSTM were able to produce more intelligible enhanced speech than recent deep learning approaches to speech enhancement, including the LSTM-IRM estimator and Xu2017. This indicates why Deep Xi-ResLSTM and Deep Xi-ResBiLSTM are more effective front-ends than SEGAN, the LSTM-IRM estimator, and Xu2017.

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